

1. Ecosystem & Its Components

- * Ecosystem: Self-sustaining functional unit of biosphere.
- * Biotic Components: Plants, animals, human, microorganisms.
- * Abiotic Components: Light, temperature, air, water, soil, minerals.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Natural Ecosystem: Forests, lakes, oceans.
- > Artificial/Man-made Ecosystem: Aquarium, crop fields, gardens.

EXAM BOOSTER & PREVIOUS BOARD QUESTIONS (BSEB / CBSE)

- Q1. Explain the main concept with suitable diagram and units.
- Q2. Write short notes on key phenomena and practical applications.
- Q3. Differentiate between primary principles and real-world examples.
- Q4. Solve numerical questions based on standard Board exam paper patterns.

2. Producers, Consumers & Decomposers

- * Producers (Autotrophs): Green plants synthesize food using sunlight.
- * Consumers (Heterotrophs): Herbivores, Carnivores, Omnivores, Parasites.
- * Decomposers (Saprotrophs): Bacteria & Fungi decompose dead organic matter.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Decomposers clean environment & replenish soil nutrients.
- > Without decomposers, nutrient cycling would stop.

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3. Food Chain & Trophic Levels

- * Food Chain: Sequential transfer of energy from producers to consumers.
- * Trophic Levels: Step or level in food chain (T1: Producers, T2: Primary Consumers, T3: Secondary, T4: Tertiary).
- * Food Web: Network of interconnected food chains in an ecosystem.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Energy flow is unidirectional (Sun -> Producers -> Consumers).
- > Food webs provide ecological stability.

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4. 10 Percent Energy Flow Law

- * Lindeman 10% Law: Only 10% of energy at a trophic level is transferred to next level.
- * 90% of energy is lost as heat in respiration & metabolic activities.
- * Number of trophic levels limited to 3-4 due to progressive energy loss.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Top carnivores receive least energy.
- > Pyramid of energy is always upright.

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5. Biological Magnification (Biomagnification)

- * Accumulation of non-biodegradable chemicals (e.g. DDT, pesticides) at higher trophic levels.
- * Chemical concentration increases progressively at each higher trophic level.
- * Human beings at top of food chain receive maximum chemical concentration.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Pesticides enter food chain through soil & water absorption by plants.
- > Non-biodegradable pollutants cannot be metabolized or excreted.

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6. Ozone Layer Depletion & Causes

- * Ozone (O₃) layer in stratosphere protects Earth from harmful UV radiation.
- * UV rays cause skin cancer, cataract & damage to immune system.
- * CFCs (Chlorofluorocarbons) from refrigerators & aerosol sprays destroy O₃.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > UNEP agreement (1987) froze CFC production globally.
- > Ozone hole observed over Antarctica.

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7. Garbage Management & Waste Types

- * Biodegradable Waste: Decomposed by microorganisms (peels, paper, leaves).
- * Non-Biodegradable Waste: Not broken down easily (plastics, glass, metals).
- * 5 R's of Waste Management: Refuse, Reduce, Reuse, Repurpose, Recycle.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Incineration, composting, and landfilling methods.
- > Plastics cause long-term soil & water pollution.

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8. Environmental Pollution & Protection

- * Air Pollution: Harmful gases, smoke, vehicle emissions.
- * Water Pollution: Industrial effluents, sewage dumping.
- * Soil Pollution: Chemical fertilizers, non-biodegradable trash.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Planting trees (Afforestation) restores ecological balance.
- > Clean energy sources: Solar, Wind, Hydroelectric power.

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9. Important Definitions & Diagrams

- * Trophic level pyramid diagrams.
- * Ozone formation reaction: $O_2 + UV \rightarrow O + O$, $O + O_2 \rightarrow O_3$.
- * Biomagnification flow chart from water/soil to human.

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Key differences between Biodegradable and Non-biodegradable waste.
- > Role of individual in environmental protection.

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10. One Page Quick Revision & Exam Booster

- * Summary of Ecosystem, 10% Law, Ozone Depletion & Waste Management.
- * Short answer & long answer board exam questions.
- * Previous 5 Years Board Questions (BSEB Class 10 Science).

MIND MAP & IMPORTANT FORMULAS / CONCEPT SUMMARY

- > Quick formula/statement review for exams.
- > Top score booster key points.

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